OM FINAL EXAM CHEAT SHEET

1. HOW TO SOLVE + DATA EXTRACTION

Queueing: 5-step workflow

Given a service system with arrivals, servers, and waits…

1. Extract: λ (arrivals/hr), tS (avg service min), σS, S (# servers)
2. Convert: μ = 60/tS. Compute ρ = λ/(S×μ). If ρ ≥ 1 → UNSTABLE
3. CVs: CA (see ID guide), CS = σS/tS
4. Lq = [ρ^√(2(S+1)) / (1−ρ)] × [(CA² + CS²) / 2]
5. Derive: Wq = Lq/λ, W = Wq + tS, L = Lq + ρS, PCE = tS/W

HOW TO IDENTIFY: Poisson vs Deterministic

"Poisson arrivals" / "exponential inter-arrivals" → CA = 1
"Random arrivals" / phone calls / walk-ins / emergencies → CA = 1
"Deterministic" / "exactly X minutes" / scheduled → CV = 0
Machines on a fixed timer / assembly line with fixed cycle → CV = 0
Problem gives σ and mean → CV = σ / mean (compute it)
Nothing said about arrivals? → assume Poisson (CA = 1)
Nothing said about service? → assume exponential (CS = 1) unless σ given
"Exponential service times" → CS = 1 (meaning σS = tS)

FINDING λ, μ, S, σS IN THE PROBLEM TEXT

λ: "receives 10 calls/hr" → λ = 10 directly
λ: "280 orders/day, 24hr" → λ = 280/24 = 11.67/hr. Always → per hour
λ: "15 customers/hr split 2 desks" → each desk gets λ = 7.5
μ: "each call takes 20 min" → μ = 60/20 = 3/hr
μ: "processes 12 customers/hr" → μ = 12 directly
S: "4 operators" / "5 employees" / "3 lines" → count of servers
σS: "std dev of 5 min" → σS = 5, so CS = 5/20 = 0.25

POOLING: what it means for your calculations

"Separate queues" / "each floor" → S independent M/M/1s, each with own λ. Compute Lq per queue, add. "Common queue" / "centralize" → one M/M/S. Same total λ, one pooled Lq.
Pooling ALWAYS wins on Wq. Idle time at one server helps the common queue. Separate → idle capacity wasted. The √(2(S+1)) exponent makes pooled Lq dramatically lower.
Exam: answer YES to pooling operationally. But add: separate queues allow VIP/express segmentation — argue both sides for Part IV.

Key: Idle capacity is NOT waste. At ρ = 83%, adding 1 server (→ 67%) cuts wait by 76%. Exponential, not linear.

Capacity: bottleneck method

Given a multi-step process with resource constraints…

1. List each resource type + processing time per unit
2. Capacity per resource = available time ÷ time per unit
3. Resources needed = ⌈demand ÷ capacity per resource⌉ (round UP)
4. Bottleneck = resource needing the MOST units to meet demand

Little's Law: WIP = TH × TT. Universal. Know any 2 → find the 3rd.

Inventory: decision tree + data extraction

Given demand, ordering costs, and uncertainty…

1. Can you reorder? YES → multi-period. NO → newsvendor
2. Review type? Continuous (VP = LT) or Periodic (VP = LT + R)?
3. Batch size: EOQ = √(2DS / (vi))
4. Buffer: SS = z × σ_1 × √VP
5. Trigger: ROP = $\bar{D}_1$ × VP + SS. Compare IP (not on-hand!) to ROP

TRANSLATING INVENTORY EXAM PROBLEMS (2024 patterns)

"Orders daily" / "replenish" → multi-period. "One season" → newsvendor
"Checks at 8 AM" / "weekly order" → periodic (VP = LT+R). "ERP real-time" → continuous (VP = LT)
"Compare suppliers" → full 6-line cost for each. Different v, S, LT → different EOQ, SS, total
"Optimal policy?" → give EOQ + SS + ROP (or S). "Annual cost?" → sum all 6 lines
"How much safety stock?" → need z, σ_1, VP. Watch VP trap!

HOW TO IDENTIFY THE SERVICE LEVEL

Explicit: "99% service level" / "SL = 95%" → use z from table
Explicit: "stockout probability of 1%" → SL = 99% → z = 2.33
Problem gives z directly: "z = 1.64" → use it, SL = 95%
Implied by context (when nothing is stated):
Medical / pharmaceutical / critical supplies → 99%
Standard consumer goods / grocery → 95%
Low-priority / cheap items → 90%

FINDING D, S, v, i, σ_1, LT, R IN PROBLEM TEXT

D: "annual demand 36,500" → D = 36,500. "100/day" → D = 36,500/yr
S (order cost): "€100 per order" / "setup cost" / "fixed delivery fee"
S is the FIXED cost per order — NOT the unit price!
v: "€500 per box" / "unit cost" / "purchase price" → unit value
i: "10% holding rate" / "carrying cost 20%" → annual rate as decimal
i given as WEEKLY? → × 52. "0.5%/week" → 0.005 × 52 = 26%/yr
i given as DAILY? → × 365. "0.03%/day" → 0.0003 × 365 = 10.95%/yr
i is ALWAYS annual in EOQ. Convert weekly/daily to annual first.
σ_1: "daily std dev = 30" → σ_1 = 30. Must match time unit of $\bar{D}_1$
LT: "lead time 7 days" / "delivery in 1 week" → LT = 7
R: "orders placed daily at 8 AM" → R = 1. "weekly" → R = 7
Continuous review → R = 0 (no gap between reviews)

⚠ EXAM TRAP: VP = LT + R in periodic review. 2024: R = 1 (daily at 8 AM). VP = 8, not 7. Affected SS, ROP, and total cost.

Supplier / policy cost comparison (2024 exam pattern)

Step-by-step: TOTAL ANNUAL COST for each supplier

1. Purchasing = D × v (usually the biggest number)
2. Ordering = (D / Q) × S (orders/yr × cost per order)
3. Holding (cycle) = (Q / 2) × v × i (avg inventory × value × rate)
4. Holding (SS) = SS × v × i (buffer × value × rate)
5. Transport = D × transport/unit (annual volume × per-unit rate)
6. Warehouse = fixed annual amount (given directly)

→ TOTAL = sum all 6 lines. Compare between suppliers. Lower wins.
→ Non-differential costs cancel in EOQ but NOT in total — always include!

Supplier A vs B decision steps

1. For EACH: EOQ from differential costs only
2. For EACH: SS from that supplier's LT (shorter → less SS → less holding)
3. For EACH: build FULL total cost using the 7 components above
4. Compare totals. Purchasing difference usually dominates — but shorter LT saves SS holding too

2. FORMULA REFERENCE

Queueing

Utilization $\rho = \frac{\lambda}{S \times \mu}$

↳ *First thing to compute. ρ ≥ 1 → unstable. Above 85% → danger zone.*

ρ	utilization (fraction, 0 to 1)
λ	arrival rate (customers/hr)
S	number of parallel servers
μ	service rate per server (customers/hr)

Avg queue length $L_q = \frac{\rho^{\sqrt{2(S+1)}}}{1-\rho} \times \frac{C_A^2 + C_S^2}{2}$

↳ *THE core formula. (How busy) × (how variable). Everything derives from Lq.*

Lq	avg number waiting in queue
CA	CV of arrivals: Poisson=1, determ=0, or σA/tA
CS	CV of service: exponential=1, determ=0, or σS/tS

Avg wait $W_q = \frac{L_q}{\lambda}$

↳ *Little's Law for the queue. Check against SLA targets.*

Total time $W = W_q + t_S$

↳ *Total customer experience: waiting in queue + being served.*

Service time $t_S = \frac{1}{\mu}$

tS = average service time per customer. μ = service rate = how many customers one server handles per hour. Find μ from the problem: "each call takes 20 min" → tS = 20 min → μ = 60/20 = 3/hr. "Processes 12 customers/hr" → μ = 12 → tS = 1/12 hr = 5 min. Always convert to match your time units (usually hours).

 # in system $L = L_q + \rho \times S$

↳ *Everyone inside. Use when cost is per-item-in-system.*

PCE $PCE = \frac{t_S}{W} = 1 - \frac{W_q}{W}$

↳ *% of time in actual service. Most services: PCE < 15%.*

Capacity

Little's Law $WIP = TH \times TT$

↳ *Universal. No assumptions. Know any 2 → find 3rd.*

WIP	work in progress (# units in system)
TH	throughput (units per time)
TT	throughput time (time per unit)

Bathtub $WIP_{end} = WIP_{start} + (Inflow - Outflow) \times Time$

↳ *When demand temporarily exceeds capacity. How backlog builds.*

Resources $Resources = \lceil \frac{Demand}{Cap\ /\ resource} \rceil$

↳ *Round UP. Resource needing the most = bottleneck.*

Inventory — EOQ (batch size)

EOQ $EOQ = \sqrt{\frac{2 \times D \times S}{v \times i}}$

↳ *Optimal batch size. Balances ordering (↓ with big Q) vs holding (↑). Flat near optimum: ±10% → ~0.5% cost penalty.*

D	annual demand (units/yr)
S	fixed cost per order (€) — NOT unit price!
v	unit value (€/unit)
i	annual holding rate (decimal). Weekly × 52, daily × 365.

TC at EOQ $TC = \sqrt{2 \times D \times S \times v \times i}$

TC = Total Cost per year at optimal Q = ordering cost + cycle holding cost (these two are equal at EOQ). It does NOT include purchasing (D×v), SS holding (SS×v×i), transport, or warehouse. For full supplier comparison, compute all 6 cost lines.

Ordering cost $\frac{D}{Q} \times S$

↳ *Annual spend on placing orders.*

Holding (cycle) $\frac{Q}{2} \times v \times i$

↳ *Avg inventory cost. At EOQ, this ≈ ordering cost.*

Holding (SS) $SS \times v \times i$

↳ *Buffer always held. Doesn't change with Q.*

Safety Stock & ROP (timing)

Safety stock $SS = z \times \sigma_1 \times \sqrt{VP}$

↳ *Buffer against uncertainty. z from table. σ_1 = per-period std dev.*

z	safety factor (99% → 2.33, 95% → 1.64, 90% → 1.28)
σ_1	std dev of ONE period's demand
VP	vulnerable period (see below)

VP (continuous) $VP = LT$

↳ *Inventory monitored in real time. Only exposed during lead time.*

VP (periodic) $VP = LT + R$

↳ *Checked every R days. Blind between reviews → longer exposure. THE #1 exam trap.*

ROP $ROP = D_1 \times VP + SS$

ROP = Reorder Point. When IP hits this, order. IP (Inventory Position) = on-hand + on-order − backorders. NOT on-hand alone — goods in transit count. On-hand only → you double-order.

Order-up-to (S) $S = D_1 \times (LT + R) + z \times \sigma_1 \times \sqrt{LT + R}$

S = Order-up-to level for periodic review. Every R days, check IP and order (S − IP). VP = LT + R is built into the formula. IP = on-hand + on-order − backorders.

Stockouts/yr $(1 - SL) \times \frac{D}{Q}$

↳ *99% SL + 36 orders → ~1 stockout every 3 years.*

z-Table (memorize 90%, 95%, 99%)

SL	z	SL	z

50%	0.00	90%	1.28
80%	0.84	95%	1.64
83.3%	0.97	99%	2.33
85%	1.04	99.9%	3.09

Three to memorize: 90% → 1.28, 95% → 1.64, 99% → 2.33.

3. EXAM-PATTERN EXAMPLES

Queueing: Call Center SLA (2024 Part III, 20 pts)

A hospital distributor gets 10 calls/hour. Each call takes 20 min to handle (σ = 5 min). They have 4 operators. The hospital demands avg wait ≤ 3 min. Does the setup meet this? If not, how many operators?
→ Check if S = 4 meets the 3-min SLA. Find minimum S.

◆ "10 calls/hr" → λ = 10. Phone calls → Poisson → CA = 1
◆ "20 min/call" → tS = 20, so μ = 60/20 = 3/hr
◆ σS given (5 min) → CS = σS/tS = 5/20 = 0.25

ρ = 10 / (4 × 3) = 83.3% — above 85% danger zone!
Lq = [0.833^√10 / 0.167] × [(1 + 0.0625) / 2]
= [0.833^3.16 / 0.167] × 0.531 = 1.81
Wq = 1.81/10 × 60 = 10.8 min → FAILS SLA ✗

S = 5: ρ = 10/(5×3) = 66.7%, Lq = 0.44, Wq = 2.6 min → PASSES ✓
One extra operator: 83% → 67%, wait drops 76%
◆ POOLING: separate queues → compute each as M/M/1. Merging → one M/M/S with same total λ. Pooled always has lower Wq. Argue: operationally optimal, but segmentation has strategic value.

Capacity: Kit Preparation (2024 Part II, 20 pts)

A distributor prepares 100 surgical kits/day across two 4-hr shifts. Each kit: picking (30 min, 1 employee), checking (10 min, 1 employee + 1 scale), sterilization (60 min, 1 machine). How many of each resource?
→ Find employees, scales, sterilization units. Identify bottleneck.

◆ "4-hour shift" → available = 240 min/shift. Each resource is independent.

Picking: 240/30 = 8 kits/shift → ⌈100/8⌉ = 13 employees
Checking: 240/10 = 24 kits/shift → ⌈100/24⌉ = 5 scales + 5 employees
Total employees = 13 + 5 = 18
Sterilization: 240/60 = 4 kits/shift → ⌈100/4⌉ = 25 machines

Bottleneck = sterilization (25 units — needs the most)
◆ Always ⌈round UP⌉. ⌈12.5⌉ = 13, never 12.

Inventory: Full Cost Comparison (2024 Part I, 20 pts)

Medcorp: 100 boxes/day (σ = 30, D = 36,500/yr). Daily review at 8 AM (R = 1). SL = 99%, i = 10%, warehouse = €2,000/yr. US: v = €500, S = €100, transport = €10/box, LT = 7. Portugal: v = €300, S = €150, transport = €7.50/box, LT = 1.
→ Full cost comparison. Which supplier wins?

◆ "Daily at 8 AM" → periodic → R = 1 → VP = LT + R
◆ z = 2.33 (99%). σ_1 = 30 (daily). S = order cost, NOT servers!
◆ Transport/box is non-differential → excluded from EOQ, included in total

US: VP = 7+1 = 8. SS = 2.33×30×√8 = 197.8. EOQ = 382
Purchasing: 36,500×500 = 18,250,000
Ordering: (36,500/382)×100 = 9,555
Hold(cycle): (382/2)×500×0.10 = 9,550
Hold(SS): 197.8×500×0.10 = 9,890
Transport: 36,500×10 = 365,000 | Warehouse: 2,000
★ TOTAL US = 18,645,995 €/yr

PT: VP = 1+1 = 2. SS = 2.33×30×√2 = 98.9. EOQ = 604
Purchasing: 36,500×300 = 10,950,000
Ordering: 9,065 | Hold(cycle): 9,060 | Hold(SS): 2,967
Transport: 273,750 | Warehouse: 2,000
★ TOTAL PT = 11,246,842 €/yr

Portugal saves ~7.4M €/yr. Lower v dominates.

4. TOP 5 EXAM TRAPS

1. **VP = LT + R in periodic review.** 2024: R = 1, VP = 8 not 7.
2. **Non-differential costs don't change EOQ.** But DO affect total cost. Always include.
3. **SS doesn't change when Q changes.** SS = f(z, σ_1, VP). Not f(Q).
4. **Compare IP to ROP, not on-hand.** IP = on-hand + on-order − backorders.
5. **ρ near 100% is catastrophic.** 83% → 67% cuts Wq by 76%.

5. PART IV STRATEGY (30-40 pts)

THE STRATEGIC TRIANGLE: Cost — Range — Service

No firm can minimize cost, max range, AND fast service. Pick 2.
Low cost + fast service → limited range (fast food)
Wide range + fast service → high cost (ER)
Low cost + wide range → slow service (Amazon)
Frame Part IV: which 2 does the firm pick? What is sacrificed?

ρ > 85%? → Stabilize first. Add capacity or reduce variability.
Pooling vs separation? → Pooling ALWAYS wins on Wq. Separate only for VIP/express. Argue both.
Which supplier? → Lower v dominates. SS savings from shorter LT are secondary. Quantify both.
Centralize? → Pools inventory (SS drops √n). But loses local relationships, speed, market knowledge.
Part IV structure? → (1) Quantify with formulas. (2) Argue BOTH sides via triangle. (3) Recommend with caveats.
Ethics/people? → Acknowledge results but flag ethical issues. Discipline, don't fire. Fix the system.